

Grade 12 Informatics Practices (065) — Worksheet - 2

Section A — Multiple Choice Questions

1 MARK EACH

1. SQL String Functions

Q1. Which SQL function is used to find the position of a substring in a string?

- a) LENGTH() b) INSTR() c) SUBSTRING() d) TRIM()

Q2. Which SQL function is used to remove leading and trailing spaces from a string?

- a) LTRIM() b) RTRIM() c) TRIM() d) LENGTH()

Q3. Which of the following SQL functions will convert the string 'hello' to 'HELLO'?

- a) LCASE('hello') b) LOWER('hello') c) UCASE('hello') d) SUBSTRING('hello')

2. SQL Math / Numeric Functions

Q4. What will be the output of the following SQL query?

```
SELECT ROUND(15.78, 0);
```

- a) 15 b) 16 c) 15.8 d) 15.78

Q5. Which SQL function will return the remainder when one number is divided by another?

- a) MOD() b) ROUND() c) POWER() d) TRIM()

Q6. What will be the output of `SELECT MOD(17, 5);` ?

- a) 3 b) 3.4 c) 2 d) 5

3. SQL Date Functions

Q7. Which SQL function is used to display the name of the day (e.g., Monday, Tuesday) from a date?

- a) DAY() b) DATE() c) DAYNAME() d) MONTHNAME()

Q8. Which SQL function returns the month name from a given date?

- a) MONTH() b) MONTHNAME() c) DATE() d) DAYNAME()

Q9. What will be the output of `SELECT DAYNAME('2026-01-26');` ?

- a) 26 b) January c) Monday d) 2026

4. SQL Aggregate Functions

Q10. Which aggregate function is used to count the total number of non-NULL values in a column?

- a) COUNT(*)
- b) COUNT(column_name)
- c) SUM(column_name)
- d) AVG(column_name)

Q11. Which of the following is NOT an aggregate function?

- a) SUM()
- b) AVG()
- c) INSTR()
- d) MAX()

Q12. What will be the output of `SELECT COUNT (*) FROM PRODUCT WHERE Qty > 20;` if the table has 5 rows and 3 rows have Qty > 20?

- a) 3
- b) 5
- c) 2
- d) 0

5. SQL Clauses (ORDER BY, GROUP BY, HAVING)

Q13. Which of the following SQL commands is used to sort rows in ascending or descending order of values in a specific column?

- a) GROUP BY
- b) ORDER BY
- c) SORT BY
- d) HAVING

Q14. Which SQL clause is used to filter groups created by the GROUP BY clause?

- a) WHERE
- b) ORDER BY
- c) HAVING
- d) DISTINCT

Q15. Which clause is used to group rows that have the same values in specified columns?

- a) ORDER BY
- b) GROUP BY
- c) HAVING
- d) WHERE

6. Keys in Database

Q16. A table has 3 candidate keys. Among these, one is chosen as the primary key. How many alternate keys does the table have?

- a) 1
- b) 2
- c) 3
- d) 4

Q17. Which key uniquely identifies each record in a table and cannot have NULL values?

- a) Alternate key
- b) Foreign key
- c) Primary key
- d) Candidate key

Q18. A key that is a candidate key but not chosen as the primary key is called:

- a) Foreign key
- b) Alternate key
- c) Composite key
- d) Super key

7. Network Topologies

Q19. In a network topology where every node is directly connected to every other node, the topology is known as:

- a) Star
- b) Bus
- c) Mesh
- d) Tree

Q20. In which topology are all nodes connected to a single central backbone cable?

- a) Star
- b) Bus
- c) Mesh
- d) Ring

Q21. In which topology are all nodes connected to a central hub or switch?

- a) Bus b) Mesh c) Star d) Tree

8. Network Devices

Q22. Which of the following is a network device used to connect two different networks and forward data packets between them?

- a) Hub b) Switch c) Router d) Repeater

Q23. What is the purpose of a modem in a computer network?

- a) To amplify signals b) To convert digital signals to analog signals and vice versa
c) To route data packets between networks d) To connect devices within a LAN

Q24. Which device is used to regenerate weak signals in a network?

- a) Hub b) Switch c) Router d) Repeater

9. Types of Networks (PAN, LAN, MAN, WAN)

Q25. Which type of network is formed when the Hyderabad campus of a company connects to its Bengaluru headquarters?

- a) PAN b) LAN c) MAN d) WAN

Q26. A network spread within a single room or a person's workspace is called:

- a) PAN b) LAN c) MAN d) WAN

Q27. A network covering a city or a metropolitan area is called:

- a) PAN b) LAN c) MAN d) WAN

Section B — Short Answer Questions 2 MARKS EACH

10. ORDER BY vs GROUP BY (Differentiate)

Q28. Differentiate between ORDER BY and GROUP BY in SQL with suitable examples.

Q29. Differentiate between WHERE and HAVING clauses in SQL with suitable examples.

Q30. Differentiate between Primary Key and Foreign Key with suitable examples.

11. PAN and LAN (Explain with Examples)

Q31. Explain the concepts of PAN and LAN with a suitable example for each.

Q32. Explain the concepts of MAN and WAN with a suitable example for each.

Q33. Differentiate between LAN and WAN on the basis of speed, coverage, and cost.

12. SQL Query Output (Aggregate Functions)

Q34. Write the output of the following SQL queries based on the table PRODUCT:

Prod_ID	Prod_Name	Price	Qty
P101	Laptop	60000	15
P102	Mouse	500	50
P103	Keyboard	800	30
P104	Monitor	12000	10
P105	Printer	15000	5

- I. `SELECT MAX(Price), MIN(Price) FROM PRODUCT;`
- II. `SELECT COUNT(*) FROM PRODUCT WHERE Qty > 20;`

Q35. Write the output of the following SQL queries based on the table STUDENT:

RollNo	Name	Marks	Grade
1	Amit	85	A
2	Priya	72	B
3	Rohan	90	A
4	Sneha	65	C
5	Vikas	78	B

- I. `SELECT AVG(Marks) FROM STUDENT;`
- II. `SELECT COUNT(*) FROM STUDENT WHERE Grade = 'A';`

Q36. Write the output of the following SQL queries based on the table EMP:

EmpID	Name	Salary	Dept
E01	Raj	50000	HR
E02	Simran	60000	IT
E03	Kabir	45000	HR
E04	Ananya	70000	IT

- I. `SELECT SUM(Salary) FROM EMP WHERE Dept = 'HR';`
- II. `SELECT MAX(Salary) FROM EMP;`

13. SQL Date Functions (Query Writing)

- Q37.** Write a SQL query to display the current date and the month name of the current date.
- Q38.** Write a SQL query to display the current date, day name, and month name of the current date.
- Q39.** Write a SQL query to display the name of the day for the date '2026-01-26'.

14. Hub vs Switch (Differentiate)

- Q40.** Explain the difference between a hub and a switch in a computer network.
- Q41.** Explain the difference between a router and a gateway in a computer network.
- Q42.** Explain the difference between a repeater and a router in a computer network.

Section C — Descriptive / Query-Based Questions

4 MARKS EACH

15. Network Topologies (Star vs Bus)

- Q43.** Differentiate between Star topology and Bus topology with suitable diagrams and explanations.
- Q44.** Differentiate between Mesh topology and Tree topology with suitable diagrams and explanations.
- Q45.** Explain any two network topologies with their advantages and disadvantages.

16. SQL String Functions (Query Writing)

- Q46.** Write SQL queries for the following:
- I. To display the position of the substring 'Science' in the string 'Computer Science'.
 - II. To display the length of the string 'Informatics Practices'.
 - III. To extract the first 5 characters from the string 'Database Management'.
- Q47.** Write SQL queries for the following:
- I. To convert the string 'database' to uppercase.
 - II. To remove leading and trailing spaces from the string ' Hello World '.
 - III. To display the first 3 characters of the string 'Programming'.
- Q48.** Write SQL queries for the following:
- I. To display the position of 'Practice' in 'Informatics Practices'.
 - II. To display the length of the string 'Computer Science'.
 - III. To extract the last 4 characters from the string 'Database Management'.

17. SQL Joins (Equi-Join, Natural Join)

- Q49.** Consider the table EMPLOYEE and DEPARTMENT given below:

Table: EMPLOYEE

Emp_ID	Emp_Name	Dept_ID	Salary	Join_Date
101	Arvind	D01	75000	2020-05-15
102	Bhavna	D02	55000	2019-08-20
103	Chetan	D03	65000	2021-01-10
104	Divya	D01	80000	2018-11-25
105	Esha	D04	60000	2022-03-01
106	Farhan	D01	90000	2020-07-12
107	Gauri	D02	48000	2021-09-05

Table: DEPARTMENT

Dept_ID	Dept_Name	Location
D01	IT	Mumbai
D02	HR	Delhi
D03	Finance	Mumbai
D04	Marketing	Chennai
D05	Sales	Bangalore

Write SQL queries for the following:

- I. To display the employee name, department name, and salary of all employees using an equi-join.
- II. To display the department name and the total salary paid to each department using natural join.
- III. To display the employee name, department name, and joining month for employees working in the 'IT' department.
- IV. To display the department name, location, and the number of employees in each department, but only for departments that have more than 2 employees.

Q50. Consider the table STUDENT and MARKS given below:

Table: STUDENT

StudentID	Name	Class	City
101	Ankit Sharma	12	Delhi
102	Priya Singh	11	Mumbai
103	Rohan Verma	12	Delhi

104	Shreya Patel	11	Chennai
105	Rehan Khan	12	Mumbai

Table: MARKS

StudentID	Subject	Score
101	Mathematics	85
101	Physics	78
102	Chemistry	88
103	Mathematics	92
104	Biology	81
105	Computer Science	93
102	Mathematics	76

Write SQL queries for the following:

- I. To display the name of students in Class 12 in ascending order of their names.
- II. To display the distinct subject names in uppercase where the score is greater than 80.
- III. To display the student name, subject, and score using an equi-join.
- IV. To display the student name and total marks for students who have appeared in more than one subject.

Q51. Consider the table PRODUCT and SUPPLIER given below:

Table: PRODUCT

Prod_ID	Prod_Name	Price	Supplier_ID
P01	Laptop	60000	S01
P02	Mouse	500	S02
P03	Keyboard	800	S01
P04	Monitor	12000	S03
P05	Printer	15000	S02

Table: SUPPLIER

Supplier_ID	Supplier_Name	City
S01	TechWorld	Mumbai
S02	DigiStore	Delhi

Write SQL queries for the following:

- I. To display the product name, supplier name, and price using an equi-join.
- II. To display the supplier name and the total price of products supplied by each supplier.
- III. To display the product name, supplier name, and city for products supplied by suppliers from Delhi.
- IV. To display the supplier name, city, and the number of products supplied, but only for suppliers who supply more than 1 product.

18. Network Devices (Repeater, Router, Gateway, Switch)

Q52. Define the following network devices and explain their functions with block diagrams:

- I. Repeater II. Router III. Gateway IV. Switch

Q53. Define the following network devices and explain their functions:

- I. Hub II. Modem III. Switch IV. Router

Q54. Explain the following network devices with their functions and diagrams:

- I. Repeater II. Gateway III. Hub IV. Modem

Section D — Long Answer Questions

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19. SQL Query Output (Joins, Aggregate, GROUP BY, HAVING)

Q55. Consider the table STUDENT and MARKS given below:

Table: STUDENT

StudentID	Name	Class	City
101	Ankit Sharma	12	Delhi
102	Priya Singh	11	Mumbai
103	Rohan Verma	12	Delhi
104	Shreya Patel	11	Chennai
105	Rehan Khan	12	Mumbai

Table: MARKS

StudentID	Subject	Score
101	Mathematics	85
101	Physics	78

102	Chemistry	88
103	Mathematics	92
104	Biology	81
105	Computer Science	93
102	Mathematics	76

Predict the output of the following SQL queries:

- I. `SELECT Name FROM STUDENT WHERE Class = '12' ORDER BY Name ASC;`
- II. `SELECT DISTINCT UCASE(Subject) FROM MARKS WHERE Score > 80;`
- III.
`SELECT STUDENT.Name, MARKS.Subject, MARKS.Score FROM STUDENT, MARKS
WHERE STUDENT.StudentID = MARKS.StudentID;`
- IV.
`SELECT STUDENT.Name, SUM(MARKS.Score) AS Total_Marks FROM STUDENT, MARKS
WHERE STUDENT.StudentID = MARKS.StudentID GROUP BY STUDENT.Name HAVING
COUNT(MARKS.Subject) > 1;`
- V.
`SELECT Subject, AVG(Score) AS Average_Score FROM MARKS GROUP BY Subject;`

Q56. Consider the table EMPLOYEE and DEPARTMENT given below:

Table: EMPLOYEE

Emp_ID	Emp_Name	Dept_ID	Salary	Join_Date
101	Arvind	D01	75000	2020-05-15
102	Bhavna	D02	55000	2019-08-20
103	Chetan	D03	65000	2021-01-10
104	Divya	D01	80000	2018-11-25
105	Esha	D04	60000	2022-03-01
106	Farhan	D01	90000	2020-07-12
107	Gauri	D02	48000	2021-09-05

Table: DEPARTMENT

Dept_ID	Dept_Name	Location
D01	IT	Mumbai
D02	HR	Delhi
D03	Finance	Mumbai

D04	Marketing	Chennai
D05	Sales	Bangalore

Predict the output of the following SQL queries:

I.

```
SELECT Emp_Name, Salary FROM EMPLOYEE WHERE Salary > 60000 ORDER BY Salary DESC;
```

II. `SELECT Dept_ID, COUNT(*) FROM EMPLOYEE GROUP BY Dept_ID;`

III.

```
SELECT EMPLOYEE.Emp_Name, DEPARTMENT.Dept_Name FROM EMPLOYEE, DEPARTMENT WHERE EMPLOYEE.Dept_ID = DEPARTMENT.Dept_ID;
```

IV.

```
SELECT DEPARTMENT.Dept_Name, SUM(EMPLOYEE.Salary) AS Total_Salary FROM EMPLOYEE, DEPARTMENT WHERE EMPLOYEE.Dept_ID = DEPARTMENT.Dept_ID GROUP BY DEPARTMENT.Dept_Name HAVING COUNT(*) > 1;
```

V. `SELECT Dept_ID, AVG(Salary) FROM EMPLOYEE GROUP BY Dept_ID;`

Q57. Consider the table PRODUCT and SUPPLIER given below:

Table: PRODUCT

Prod_ID	Prod_Name	Price	Supplier_ID
P01	Laptop	60000	S01
P02	Mouse	500	S02
P03	Keyboard	800	S01
P04	Monitor	12000	S03
P05	Printer	15000	S02

Table: SUPPLIER

Supplier_ID	Supplier_Name	City
S01	TechWorld	Mumbai
S02	DigiStore	Delhi
S03	ComputerHub	Chennai

Predict the output of the following SQL queries:

I.

```
SELECT Prod_Name, Price FROM PRODUCT WHERE Price > 10000 ORDER BY Price ASC;
```

II. `SELECT Supplier_ID, COUNT(*) FROM PRODUCT GROUP BY Supplier_ID;`

III.

```
SELECT PRODUCT.Prod_Name, SUPPLIER.Supplier_Name FROM PRODUCT, SUPPLIER  
WHERE PRODUCT.Supplier_ID = SUPPLIER.Supplier_ID;
```

IV.

```
SELECT SUPPLIER.Supplier_Name, SUM(PRODUCT.Price) AS Total_Price FROM  
PRODUCT, SUPPLIER WHERE PRODUCT.Supplier_ID = SUPPLIER.Supplier_ID GROUP  
BY SUPPLIER.Supplier_Name HAVING COUNT(*) > 1;
```

V. `SELECT Supplier_ID, AVG(Price) FROM PRODUCT GROUP BY Supplier_ID;`

20. Network Design & Cabling (Case Study)

Q58. XYZ Tech Solutions is setting up its new office in Hyderabad with four departments: HR, Finance, IT, and Logistics. The distances between departments and the number of computers in each department are given below:

Distances between departments:

From	To	Distance (in meters)
HR	Finance	50
HR	IT	175
HR	Logistics	90
Finance	IT	60
Finance	Logistics	70
IT	Logistics	60

Number of Computers in each department:

Department	Number of Computers
HR	60
Finance	40
IT	90
Logistics	30

- I. Suggest the most suitable department to install the server. Justify your reason.
- II. Draw a suitable cable layout of connections between the departments in the campus.
- III. Which hardware device will you suggest to connect all the computers within each department?
- IV. Which network topology is most suitable for connecting the departments in this campus? Justify your answer.

Q59. Shiva Corporation is planning a new campus in Pune while maintaining its headquarters in Mumbai. The campus will have four buildings: Administration, Sales, Development, and Support. As a

network expert, you are tasked with proposing the best network solutions for their needs based on the following:

Distances between buildings:

From	To	Distance (in meters)
Administration	Sales	60
Administration	Development	90
Administration	Support	120
Sales	Development	50
Sales	Support	70
Development	Support	45

Number of Computers in each building:

Building	Number of Computers
Administration	120
Sales	40
Development	70
Support	25
Mumbai HQ	150

- I. Suggest the most suitable building in the Pune campus to install the server. Justify your reason.
- II. Suggest and draw a cable layout of connections between the buildings inside the Pune campus.
- III. Which hardware device will you suggest to connect all the computers within each building?
- IV. The organization plans to provide a high-speed wired link with its Mumbai headquarters. Which type of cable will be most suitable for this job?
- V. Which type of network (PAN, LAN, MAN, or WAN) will be formed when connecting the Pune campus to the Mumbai headquarters?

Q60. ABC Ltd. is setting up a new branch in Chennai with four departments: Production, Marketing, Accounts, and HR. The distances between departments and the number of computers are given below:

Distances between departments:

From	To	Distance (in meters)
Production	Marketing	80
Production	Accounts	100
Production	HR	120

Marketing	Accounts	60
Marketing	HR	90
Accounts	HR	50

Number of Computers:

Department	Number of Computers
Production	50
Marketing	35
Accounts	25
HR	20

- I. Suggest the most suitable department to install the server. Justify your reason.
- II. Draw a suitable cable layout of connections between the departments.
- III. Which hardware device will you suggest to connect all the computers within each department?
- IV. Which network topology is most suitable for connecting the departments? Justify your answer.
- V. Which type of network will be formed if this branch connects to the head office in Delhi?